# Runtime comparison | Intel Core i9-14900K / Linux

Independent release-candidate evaluation on 25 October 2026

Seconds | median of three runs

| Problem | $n$ | $\gamma$ | **LSQN** | VI | PI | MPI-4 | MPI-20 | QPI-U | QPI-BT | QPI-S | iGMRES-PI | AA-I-S-5 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Queue | 512 | 0.99 | 0.0413 | 0.0734 | **0.0062** | 0.3089 | 0.0672 | 0.1082 | 0.0960 | 1.3676 | 0.1103 | 0.0427 |
| | | 0.999 | 0.0479 | 0.7154 | **0.0051** | 3.0761 | 0.6798 | 0.5364 | 1.4866 | 10.7680 | TO | F(0/3) |
| | 1024 | 0.99 | 0.1590 | 0.0928 | **0.0077** | 0.3211 | 0.0718 | 0.1278 | 0.1162 | 7.3198 | 0.1256 | 0.0552 |
| | | 0.999 | 0.2696 | 0.9316 | **0.0051** | 3.3093 | 0.7393 | 0.9228 | 2.0167 | TO | TO | 0.3936 |
| | 2048 | 0.99 | 0.3735 | 0.1379 | **0.0085** | 0.3682 | 0.0842 | 0.1704 | 0.1545 | 26.3219 | 0.1379 | 0.0776 |
| | | 0.999 | 6.5113 | 1.3878 | **0.0073** | 3.7483 | 0.8636 | 1.5837 | 2.2774 | TO | TO | 0.6538 |
| | 4096 | 0.99 | 0.6700 | 0.2325 | **0.0120** | 0.4615 | 0.1096 | 0.2658 | 0.2444 | TO | 0.1883 | 0.1205 |
| | | 0.999 | 27.8876 | 2.3267 | **0.0103** | 4.6816 | 1.0942 | 2.7690 | 2.5260 | TO | TO | 1.5413 |
| Cluster | 512 | 0.99 | **0.0056** | 0.2322 | 0.0098 | 0.1472 | 0.0625 | 0.1014 | 0.0974 | 0.0477 | 0.0089 | 0.0090 |
| | | 0.999 | **0.0060** | 2.2990 | 0.0146 | 1.4576 | 0.6103 | 0.1898 | 0.1781 | 0.0513 | 0.0111 | 0.0181 |
| | 1024 | 0.99 | **0.0215** | 1.4990 | 0.0938 | 1.0609 | 0.3467 | 0.5081 | 0.5020 | 0.2419 | 0.0354 | 0.0316 |
| | | 0.999 | **0.0225** | 15.4677 | 0.0936 | 10.6944 | 3.4448 | 0.9523 | 0.9424 | 0.3131 | 0.0377 | 0.0862 |
| | 2048 | 0.99 | **0.0938** | 6.4847 | 0.7048 | 5.9942 | 2.2436 | 2.2153 | 2.2206 | 0.7413 | 0.1916 | 0.1274 |
| | | 0.999 | **0.1100** | TO | 0.7173 | TO | 22.7978 | 4.0976 | 4.1003 | 3.4686 | 0.2132 | 0.2905 |
| | 4096 | 0.99 | **0.3837** | 27.3391 | 5.1741 | 26.6504 | 10.5957 | 8.9088 | 8.9581 | 2.8831 | 0.7862 | 0.6159 |
| | | 0.999 | **0.4742** | TO | 5.1796 | TO | TO | 16.8628 | 16.9474 | 3.1926 | 0.8691 | 0.9481 |
| Garnet | 512 | 0.99 | 0.0025 | 0.0761 | 0.0212 | 0.3278 | 0.0725 | 0.0021 | **0.0018** | 0.0243 | 0.0074 | 0.0041 |
| | | 0.999 | 0.0026 | 0.7630 | 0.0210 | 3.2529 | 0.7238 | 0.0021 | **0.0018** | 0.0235 | 0.0099 | 0.0038 |
| | 1024 | 0.99 | 0.0032 | 0.0979 | 0.1530 | 0.3659 | 0.0834 | 0.0026 | **0.0024** | 0.1257 | 0.0088 | 0.0049 |
| | | 0.999 | 0.0033 | 0.9790 | 0.1573 | 3.6586 | 0.8248 | 0.0027 | **0.0024** | 0.1264 | 0.0105 | 0.0049 |
| | 2048 | 0.99 | 0.0051 | 0.1521 | 1.1825 | 0.4671 | 0.1094 | 0.0038 | **0.0035** | 0.6505 | 0.0106 | 0.0078 |
| | | 0.999 | 0.0052 | 1.5269 | 1.2061 | 4.6978 | 1.0758 | 0.0038 | **0.0034** | 0.6583 | 0.0122 | 0.0075 |
| | 4096 | 0.99 | 0.0082 | 0.2895 | 9.1792 | 0.6550 | 0.1547 | 0.0062 | **0.0056** | 2.9895 | 0.0134 | 0.0123 |
| | | 0.999 | 0.0085 | 2.9042 | 9.1339 | 6.5696 | 1.5377 | 0.0061 | **0.0057** | 2.9675 | 0.0147 | 0.0123 |
| Latent | 512 | 0.99 | 0.0026 | 0.2431 | 0.0098 | 0.1497 | 0.0625 | 0.0016 | **0.0013** | 0.0088 | 0.0058 | 0.0021 |
| | | 0.999 | 0.0026 | 2.3823 | 0.0102 | 1.4663 | 0.6313 | 0.0017 | **0.0013** | 0.0086 | 0.0059 | 0.0018 |
| | 1024 | 0.99 | 0.0093 | 1.4503 | 0.0628 | 1.0727 | 0.3502 | 0.0052 | **0.0051** | 0.0306 | 0.0256 | 0.0069 |
| | | 0.999 | 0.0092 | 14.7927 | 0.0639 | 10.5693 | 3.4348 | 0.0052 | **0.0051** | 0.0302 | 0.0214 | 0.0070 |
| | 2048 | 0.99 | 0.0344 | 6.3964 | 0.4796 | 5.8761 | 2.1851 | 0.0195 | **0.0192** | 0.1354 | 0.1047 | 0.0274 |
| | | 0.999 | 0.0312 | TO | 0.4842 | 59.3720 | 22.2991 | 0.0197 | **0.0193** | 0.1353 | 0.1058 | 0.0272 |
| | 4096 | 0.99 | 0.1403 | 28.1215 | 3.4494 | 24.9242 | 10.3985 | **0.0798** | 0.0799 | 0.5832 | 0.5619 | 0.1123 |
| | | 0.999 | 0.1411 | TO | 3.4724 | TO | TO | 0.0815 | **0.0801** | 0.5850 | 0.4313 | 0.1283 |
| Mean rank (lower is better) | | | 3.5000 | 8.1094 | 5.0938 | 8.8281 | 6.4375 | 4.1250 | 3.6875 | 6.9531 | 4.9688 | 3.2969 |
| Overall rank | | | 2 | 9 | 6 | 10 | 7 | 4 | 3 | 8 | 5 | 1 |
| Successful instances | | | 32/32 | 28/32 | 32/32 | 29/32 | 30/32 | 32/32 | 32/32 | 28/32 | 28/32 | 31/32 |

Entries are median seconds from three runs. Bold marks the lowest successful time in each row, using unrounded values. Success requires all three runs to satisfy the independent residual check (infinity norm at most 1e-8). TO: all three runs timed out; F(s/3): only s runs succeeded. Per-run limits: 30 s for n at most 1024; 60 s otherwise. Ranks use unrounded times; unsuccessful methods tie below successful ones.

This separate Linux run used the recorded NumPy Prescott / SciPy Haswell BLAS dispatch configuration. Ranks are descriptive. These measurements are not an isolated CPU-performance comparison; see the archive metadata.